University Enterprise Network-2 — Cisco Packet Tracer

Project Overview

Designed and configured a university enterprise network using Cisco Packet Tracer to practise switching, routing, DHCP services and Layer 2 security.

Network Design

- Student VLAN 10 — `10.0.10.0/24`
- Staff VLAN 20 — `10.0.20.0/24`
- Admin VLAN 30 — `10.0.30.0/24`
- IT VLAN 40 — `10.0.40.0/24`
- Server VLAN 70 — `10.0.70.0/24`
- Management VLAN 99 — `10.0.99.0/24`

Technologies and Configuration

- Cisco Catalyst access switches and multilayer core switch
- VLANs and 802.1Q trunking
- Inter-VLAN routing using SVIs
- DHCP server and DHCP relay using `ip helper-address`
- Static routing between the core, edge router and ISP router
- PAT (NAT overload) for internal networks
- DHCP Snooping to help prevent rogue DHCP servers
- Port Security with sticky MAC address learning
- PortFast and BPDU Guard on end-device ports

Testing and Verification

- Verified DHCP address assignment
- Tested inter-VLAN connectivity
- Tested connectivity between the core switch, edge router and ISP router
- Used Cisco IOS show commands to verify network configuration

Files

- `University-Network.pkt` — Cisco Packet Tracer project
- `screenshots/` — Network topology and verification evidence

Learning Outcomes

Practised enterprise network design, VLAN configuration, routing, DHCP troubleshooting and Layer 2 security.

Note: Include only features that are configured and successfully tested in the submitted Packet Tracer file.